KHAVIN K J

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Summary

Electronics and Communication Engineering graduate with strong proficiency in Java, SQL, HTML, CSS, and JavaScript and a solid foundation in data structures, algorithms, and object-oriented programming. Experienced in developing real-time, performance-focused applications, including a computer vision–based attendance system. Skilled in designing and implementing efficient, scalable software solutions with a focus on clean code, optimization, and practical problem-solving. Seeking entry-level software engineering roles.

Education

Sathyabama Institute of Science and Technology, Chennai — **2022 – 2026**
Bachelor of Engineering – Electronics and Communication Engineering — CGPA: **9.23** / 10.0

AMG Matric Higher Secondary School — HSC: **84%** (**2022**) | SSLC: **96%** (**2020**)

Technical Skills

Programming Languages: Java, SQL, HTML5, CSS3, JavaScript

Core CS Concepts: Data Structures, Algorithms, Object-Oriented Programming, Java Collections Framework

Database: MySQL – schema design, query optimization, relational database management.

Developer Tools: VS Code,Arduino IDE

Experience

ThinkSemi Infotech Pvt Ltd — **Jan 2024**
Chennai, Tamil Nadu

- Applied systematic debugging methodology to identify and resolve faults across 10+ hardware prototypes, reducing testing time by 20%.
- Executed structured functional validation and test procedures to verify system performance against specifications.
- Documented test cases and defect logs, improving issue traceability and streamlining resolution workflows.
- Collaborated cross-functionally to identify defects and prioritize fixes, mirroring agile software development workflows.

Projects

Face Recognition-Based Attendance System (FaceLume) — **Oct 2024**

- Developed a real-time attendance system in Python using OpenCV, YOLOv5, and RetinaFace for face detection and recognition.
- Integrated anti-spoofing techniques to strengthen system security and prevent unauthorized access.
- Designed and maintained a structured database for efficient storage and retrieval of attendance records.
- Optimized the recognition pipeline for 30+ users, reducing latency and improving system throughput.
- Achieved 90% recognition accuracy under controlled conditions through model tuning and image preprocessing.

Disney+ Hotstar Clone Frontend Project — **Sep 2025**

- Built a Disney+ Hotstar-inspired web interface using HTML, CSS, and JavaScript to replicate core UI components.
- Designed structured layouts with navigation bars, movie sections, and interactive thumbnail cards using Flexbox.
- Implemented dynamic scrolling sliders and basic interactivity with JavaScript for improved user experience.
- Focused on clean UI design and smooth content browsing similar to real-world streaming platforms.

Design of Low Power,High Performance Hybrid Adder Circuit — **Sep 2025 – Apr 2026**

- Engineered a 1-bit hybrid full adder in 90 nm CMOS using PTL (XOR-XNOR) and TG logic (sum and carry).
- Developed 13T (low-power) and 15T (improved signal integrity) variants.
- Achieved **44.94%** power reduction at 0.7V compared to existing 10T adders.
- Validated performance across 10 MHz–1 GHz, ensuring scalability for high-speed VLSI systems.
- Evaluated power and voltage swing for efficiency in deep submicron applications.

Papers Published

- “A Cloud-Based Smart Energy Analyser Using IoT” — ICDSAAI, 2025.
- “Design of Low-Power, High-Performance Hybrid Full Adder Circuit” — RMKMATE 2026.